

Balkan Romance Sound Changes

User: [File: Pardess_From_Latin_to_Romanian_Phonology.pdf]

Could you extract the sound changes from Latin to Romanian in relative chronological order from this document, writing them in SPE style?

use.ai: Here is the full ordered list of sound changes extracted from Pardess (1990), expressed in SPE-style distinctive feature notation, organized by historical period.

Balkan Romance Period (1st–6th centuries A.D.)

R1 — Loss of [h]

\$\$ [+\\text{glottal}] \\rightarrow \\emptyset \$\$

\$\$ h \\rightarrow \\emptyset \$\$

R2A — Monophthongization of AE

\$\$ [+\\text{low}, -\\text{back}, -\\text{high}] \\rightarrow [+\\text{tense}, +\\text{long}] ; / ; _ ; [+\\text{high}, -\\text{back}, -\\text{syl}] \$\$

\$\$ ay \\rightarrow e{:} \$\$

R2B — Monophthongization of OE

\$\$ oy \\rightarrow e{:} \$\$

R3A — Reduction of word-final [m]

\$\$ [+\\text{nas}, +\\text{ant}, -\\text{cor}] \\rightarrow \\begin{cases} [+\\text{cor}] & \\& / _ ; \\# ; \\text{(monosyllables)} \\end{cases} \\emptyset & / _ ; \\# ; \\text{(polysyllables)} \\end{cases} \$\$

R3B — Loss of [n] before fricative

\$\$ \\begin{bmatrix} +\\text{nas} \\ +\\text{son} \\ \\end{bmatrix} \\rightarrow \\emptyset ; / ; _ ; \\begin{bmatrix} +\\text{cont} \\ -\\text{son} \\ \\end{bmatrix} \$\$

(with compensatory lengthening of the preceding vowel)

R4A — Syncope

$$\begin{bmatrix} \text{long} \\ \text{acc} \end{bmatrix} \rightarrow \emptyset ; / V ; C_i _ ; C_j ; V$$

where C_i or C_j (or both) $= [+son, +cons]$

R4B — Dissimilation [tɫ] → [kɫ]

$$\begin{bmatrix} \text{cont} \\ +\text{ant} \\ +\text{cor} \end{bmatrix} \rightarrow \begin{bmatrix} \text{ant} \\ -\text{cor} \end{bmatrix} ; / _ ; [+lat]$$

$$t \rightarrow k ; / _ ; l$$

R5A — Deletion of word-final oral/nasal stops

$$\begin{bmatrix} \text{cont} \\ -\text{son} \end{bmatrix} \rightarrow \emptyset ; / _ ; \#$$

R5B — Paragodic [e] after word-final [ɫ]

$$\emptyset \rightarrow e ; / [+lat] ; _ ; \#$$

R5C — Metathesis of word-final [r]

$$V ; r ; \# \rightarrow r ; V ; \#$$

R6 — Fronting of [w] to [β]

$$\begin{bmatrix} \text{cons} \\ -\text{syl} \\ +\text{back} \end{bmatrix} \rightarrow \begin{bmatrix} \text{cont} \\ +\text{ant} \\ -\text{cor} \end{bmatrix}$$

$$w \rightarrow \beta$$

R7 — Fortition of syllable-initial [j]

$$\begin{bmatrix} \text{cons} \\ -\text{syl} \\ -\text{back} \end{bmatrix} \rightarrow \begin{bmatrix} \text{cons} \\ +\text{del;rel} \end{bmatrix} ; / ; \$; _$$

$$j \rightarrow dj ; / ; \$ _$$

R8 — Simplification of [–son][+strid][–son] cluster

\$\$ [-\text{son}] \rightarrow \emptyset ; / ; _ ; [+ \text{strid}] ; [-\text{son}] \$\$

R9 — Deletion of [k] in [nkt]

\$\$ \begin{bmatrix} -\text{cont} \backslash -\text{ant} \backslash -\text{cor} \end{bmatrix} \rightarrow \emptyset ; / ; [+ \text{nas}] ; _ ; \begin{bmatrix} -\text{cont} \backslash +\text{ant} \backslash +\text{cor} \end{bmatrix} \$\$

\$\$ k \rightarrow \emptyset ; / ; n _ t \$\$

R10A — Sporadic assimilation [r] → [s] / _ s

\$\$ r \rightarrow s ; / ; _ ; s \quad \text{(sporadic)} \$\$

R10B — Assimilation [p] → [s] / _ s

\$\$ p \rightarrow s ; / ; _ ; s \$\$

R10Ca — Partial assimilation of velars after stressed vowel

\$\$ \begin{bmatrix} -\text{cont} \backslash -\text{ant} \backslash -\text{cor} \end{bmatrix} \rightarrow [+ \text{ant}] ; / ; \begin{bmatrix} V \backslash +\text{acc} \end{bmatrix} _ ; \begin{bmatrix} -\text{cont} \backslash +\text{ant} \backslash +\text{cor} \end{bmatrix} \end{bmatrix} \$\$

\$\$ ks \rightarrow ps, \quad kt \rightarrow pt, \quad gn \rightarrow mn \$\$

R10Cb — Total assimilation of velars before stress

\$\$ \begin{bmatrix} -\text{cont} \backslash -\text{ant} \backslash -\text{cor} \end{bmatrix} \rightarrow \begin{bmatrix} + \text{ant} \backslash +\text{cor} \end{bmatrix} ; / ; \begin{bmatrix} V \backslash -\text{acc} \end{bmatrix} _ ; \begin{bmatrix} +\text{ant} \backslash +\text{cor} \end{bmatrix} \end{bmatrix} \$\$

\$\$ ks \rightarrow ss, \quad gn \rightarrow nn \$\$

R11 — Labialization of [kw], [gw] before [a]

\$\$ \begin{bmatrix} -\text{cont} \backslash -\text{ant} \backslash -\text{cor} \end{bmatrix} \rightarrow [+ \text{ant}] ; / ; _ ; wa \$\$

\$\$ kwa \rightarrow pwa, \quad gwa \rightarrow bwa \$\$

R12 — Prothesis: [ʃsC] → [ʃisC]

\$\$ \emptyset \rightarrow i ; / ; \# _ ; \begin{bmatrix} +\text{strid} \backslash +\text{ant} \backslash +\text{cor} \end{bmatrix} \begin{bmatrix} -\text{son} \backslash -\text{back} \end{bmatrix} \$\$

R13A — Lengthening of second vowel in hiatus + stress shift

\$\$ \begin{bmatrix} -\text{low} \backslash -\text{acc} \end{bmatrix} \rightarrow \begin{bmatrix} +\text{long} \backslash +\text{acc} \end{bmatrix} ; / ; V ; _ \$\$

R13B — Raising of short unstressed [o], [e] adjacent to vowel

\$\$ \begin{bmatrix} -\text{high} \backslash -\text{long} \backslash -\text{acc} \end{bmatrix} \rightarrow [+ \text{high}] ; / !! / ; V \$\$

\$\$ o \rightarrow u, \quad e \rightarrow i ; / !! / ; V \$\$

R13C — Contraction of identical adjacent vowels

\$\$ V_i ; V_i \rightarrow V_i \$\$

R13D — Loss of syllabicity of short unstressed high vowels adjacent to vowel

\$\$ \begin{bmatrix} +\text{high} \backslash -\text{long} \backslash -\text{acc} \end{bmatrix} \rightarrow [-\text{syl}] ; / !! / ; V \$\$

\$\$ i \rightarrow j, \quad u \rightarrow w ; / !! / ; V \$\$

R13E — Laxing of stressed [e]

\$\$ \begin{bmatrix} V \backslash -\text{long} \backslash -\text{high} \backslash -\text{back} \backslash +\text{acc} \end{bmatrix} \rightarrow [-\text{tense}] \$\$

\$\$ e \rightarrow \text{varepsilon} \$\$

R13F — Lengthening of word-final [–back +high] (before C) #

\$\$ \begin{bmatrix} +\text{high} \backslash -\text{back} \end{bmatrix} \rightarrow [+ \text{long}] ; / ; _ ; (C) ; \# \$\$

R13G — Lowering of [i]

$$\begin{matrix} V \\ \text{long} \end{matrix} + \begin{matrix} \text{high} \\ \text{back} \end{matrix} \rightarrow \begin{matrix} \text{high} \end{matrix}$$

$$i \rightarrow e$$

R14 — Apheresis: loss of unstressed initial [e] before [s/z] + C

$$e \rightarrow \emptyset ; / ; \# _ ; \begin{matrix} \text{cont} \\ \text{ant} \end{matrix} \begin{matrix} \text{cor} \\ \text{strid} \end{matrix} \begin{matrix} \text{son} \end{matrix}$$

R15 — Vowel contraction in hiatus

$$\begin{matrix} V \\ \text{long} \end{matrix} \alpha \begin{matrix} \text{back} \\ \text{high} \end{matrix} \rightarrow \emptyset ; / ; _ ; \begin{matrix} V \\ \text{long} \end{matrix} \begin{matrix} \text{acc} \\ \alpha \end{matrix} \begin{matrix} \text{back} \\ \text{high} \end{matrix}$$

(two adjacent vowels of similar height contract to one long vowel)

R16 — Loss of vowel length

$$[V] \rightarrow [-\text{long}]$$

R17 — Deletion of [w] before vowel

$$\begin{matrix} \text{cons} \\ \text{syl} \end{matrix} \begin{matrix} \text{back} \end{matrix} \rightarrow \emptyset ; / ; _ ; V$$

$$w \rightarrow \emptyset ; / ; _ ; V$$

R18A — Gemination of stop before [j] after vowel

$$\begin{matrix} \text{son} \\ \text{cont} \end{matrix} \rightarrow \begin{matrix} \text{son} \\ \text{cont} \end{matrix} \begin{matrix} \text{ant} \\ \text{cor} \end{matrix} \begin{matrix} \text{del;rel} \end{matrix} ; / ; V _ ; j$$

$$C \rightarrow CC ; / ; V _ j$$

R18B — Affrication of alveolar stops before [j]

$$\begin{matrix} \text{son} \\ \text{cont} \end{matrix} \begin{matrix} \text{ant} \\ \text{cor} \end{matrix} \begin{matrix} \text{del;rel} \end{matrix} \rightarrow \begin{matrix} \text{ant} \\ \text{del;rel} \end{matrix} ; / ; _ ; j$$

$$t \rightarrow \hat{c}, \quad d \rightarrow \hat{j} ; / ; _ ; j$$

R18C — Affrication of velars before [j]

\$\$ \begin{bmatrix} -\text{cont} \setminus -\text{ant} \setminus -\text{cor} \setminus -\text{del;rel} \end{bmatrix} \rightarrow \begin{bmatrix} +\text{cor} \setminus +\text{del;rel} \end{bmatrix} ; / ; _ ; j \$\$

\$\$ k \rightarrow \hat{c}, \quad g \rightarrow \hat{j} ; / ; _ ; j \$\$

R18D — Absorption of [j] after affricate

\$\$ j \rightarrow \emptyset ; / ; \begin{bmatrix} -\text{son} \setminus -\text{cont} \setminus +\text{cor} \setminus +\text{del;rel} \end{bmatrix} _ \$\$

R18E — Assimilation of stop to following affricate

\$\$ C_i ; C_j \rightarrow C_j ; C_j \quad \text{where } C_j = \begin{bmatrix} -\text{cont} \setminus -\text{ant} \setminus +\text{cor} \setminus +\text{del;rel} \end{bmatrix} \$\$

R19 — Depalatalization of affricates after stressed vowel

\$\$ \begin{bmatrix} -\text{ant} \setminus +\text{cor} \setminus +\text{del;rel} \end{bmatrix} \rightarrow [+ \text{ant}] ; / ; \begin{bmatrix} V \setminus +\text{acc} \end{bmatrix} _ \$\$

\$\$ \hat{c} \rightarrow ts, \quad \hat{j} \rightarrow dz ; / ; V_ \{ \text{acc} \} _ \$\$

R20 — Attraction of glide (metathesis)

\$\$ \begin{bmatrix} V \setminus +\text{acc} \end{bmatrix} ; C^0 ; \begin{bmatrix} +\text{son} \setminus -\text{cons} \setminus -\text{syl} \end{bmatrix} \rightarrow \begin{bmatrix} V \setminus +\text{acc} \end{bmatrix} \begin{bmatrix} +\text{son} \setminus -\text{cons} \setminus -\text{syl} \end{bmatrix} C^0 \$\$

R21 — Spirantization of intervocalic [b]

\$\$ \begin{bmatrix} -\text{cont} \setminus +\text{ant} \setminus -\text{cor} \setminus +\text{voice} \end{bmatrix} \rightarrow [+ \text{cont}] ; / ; V _ V \$\$

\$\$ b \rightarrow \beta ; / ; V _ V \$\$

R22A — Word-initial [β] → [v]

\$\$ \beta \rightarrow v ; / ; \# _ \$\$

R22B — Occlusivization of [β] after consonant

$$\beta \rightarrow b \text{ ; / ; } C _$$

R22C — [β] → [w] before resonant

$$\beta \rightarrow w \text{ ; / ; } _ \begin{bmatrix} +\text{son} \\ +\text{cons} \end{bmatrix}$$

R22D — Sporadic [β] → [g] in the environment of [u]

$$\beta \rightarrow g \text{ ; / ! / ; } u \quad \text{(sporadic)}$$

R22E — [β] → ∅ elsewhere

$$\beta \rightarrow \emptyset$$

R23 — Degemination (except [ll], [nn], [rr])

$$C_i \text{ ; } C_i \rightarrow C_i \quad \text{where } C_i \neq \begin{bmatrix} +\text{ant} \\ +\text{cor} \\ +\text{son} \\ +\text{cons} \end{bmatrix}$$

R24 — Affrication of velars before front vowel

$$\begin{bmatrix} -\text{cont} \\ -\text{ant} \\ -\text{cor} \\ -\text{del;rel} \end{bmatrix} \rightarrow \begin{bmatrix} +\text{cor} \\ +\text{del;rel} \end{bmatrix} \text{ ; / ; } _ \begin{bmatrix} V \\ -\text{back} \end{bmatrix}$$

$$k \rightarrow \hat{c}, \quad g \rightarrow \hat{j} \text{ ; / ; } _ \text{ ; e, i}$$

R25A — Final [s] → [j] in monosyllables

$$s \rightarrow h \rightarrow \text{varsigma} \rightarrow j \text{ ; / ; } _ \text{ ; } \# \quad \text{(monosyllables)}$$

R25B — Loss of final [s] in polysyllables

$$s \rightarrow \emptyset \text{ ; / ; } _ \text{ ; } \#$$

R25C — Generalization of word-final [i] as 2sg marker

(morphological analogical spread of the 4th conjugation 2sg ending [-i])

Common Romanian Period (6th–11th centuries A.D.)

R26A — [rr] → alveolar trill

\$\$ rr \rightarrow [+ \text{rep}] \$\$

R26B — Word-initial [r] → [rr]

\$\$ r \rightarrow [+ \text{rep}] ; / ; \# _ \$\$

R27A — Tensing of [ɛ] before nasal + oral consonant

\$\$ \begin{bmatrix} V \\ - \text{tense} \end{bmatrix} \rightarrow [+ \text{tense}] ; / ; _ ; [+ \text{nas}] \\ \begin{bmatrix} C \\ - \text{nas} \end{bmatrix} \$\$

(bleeds R27B in this environment)

R27B — Diphthongization of lax [ɛ]

\$\$ \begin{bmatrix} V \\ - \text{tense} \end{bmatrix} \rightarrow \begin{bmatrix} + \text{son} \\ - \text{cons} \\ - \text{syl} \end{bmatrix} V ; / ; _ \$\$

\$\$ \text{varepsilon} \rightarrow j \text{varepsilon} \$\$

R28 — Unstressed [o] → [u]

\$\$ \begin{bmatrix} V \\ + \text{back} \\ - \text{high} \\ - \text{low} \\ - \text{acc} \end{bmatrix} \rightarrow [+ \text{high}] \$\$

\$\$ o \rightarrow u ; / ; [- \text{acc}] \$\$

R29A — Loss of stress on postposed article forms of *ellu*

\$\$ e \rightarrow [- \text{acc}] ; / ; \# _ \quad \text{(in postposed article only)} \$\$

R29B — Apheresis and enclisis of postposed article

\$\$ e \rightarrow \emptyset ; / ; \# _ \quad \text{(in postposed article only)} \$\$

R30 — Preyodization of word-initial [e]

\$\$ \emptyset \rightarrow j ; / ; \# _ ; e \quad \text{(in certain words)} \$\$

R31A — Metaphony: stressed mid-vowels laxed before [a] or [e] in following syllable

\$\$ \begin{bmatrix} V \setminus \text{high} \setminus \text{low} \end{bmatrix} + \text{acc} \rightarrow [-\text{tense}] ; / ; _ ; C^0 \begin{bmatrix} V \setminus \text{high} \setminus \text{round} \end{bmatrix} \$\$

(except where $C^0 = n$ or mC)

R31B — Diphthongization of lax mid-vowels

\$\$ \begin{bmatrix} V \setminus \text{tense} \end{bmatrix} \rightarrow \begin{bmatrix} V \setminus \text{tense} \end{bmatrix} \begin{bmatrix} +\text{low} \end{bmatrix}, \quad \text{then} \quad \begin{bmatrix} V \setminus \text{tense} \end{bmatrix} \rightarrow [-\text{syl}] \$\$

\$\$ \text{varepsilon} \rightarrow ea, \quad \text{mathopen{o}} \rightarrow oa \$\$

R32 — Stress shift and desyllabification in [éa] sequences

\$\$ \begin{bmatrix} e \setminus +\text{acc} \end{bmatrix} a \rightarrow e \begin{bmatrix} a \setminus +\text{acc} \end{bmatrix} \end{bmatrix} \$\$

\$\$ \acute{e}a \rightarrow e\acute{a} ; \text{(with } e \rightarrow [-\text{syl}]\text{)} \$\$

R33 — Raising of [a] → [ə] and [o] → [u] before non-geminate nasal

\$\$ \begin{bmatrix} V \setminus +\text{low} \end{bmatrix} \rightarrow [-\text{low}] ; / ; _ ; [+ \text{nas}, -\text{rep}] \$\$

\$\$ \begin{bmatrix} V \setminus +\text{back} \setminus \text{high} \end{bmatrix} \rightarrow [+ \text{high}] ; / ; _ ; [+ \text{nas}, -\text{rep}] \$\$

R34 — Rhotacism: [l] → [r] between vowels

\$\$ \begin{bmatrix} +\text{lat} \setminus \text{rep} \end{bmatrix} \rightarrow [-\text{lat}] ; / ; V _ V \$\$

\$\$ l \rightarrow r ; / ; V _ V \$\$

R35A — [ll] → [w] before unstressed [a]

\$\$ \begin{bmatrix} +\text{lat} \setminus +\text{rep} \end{bmatrix} \rightarrow \begin{bmatrix} -\text{cons} \setminus \text{syl} \setminus +\text{back} \end{bmatrix} ; / ; _ ; \begin{bmatrix} a \setminus \text{acc} \end{bmatrix} \$\$

\$\$ \text{II} \rightarrow w ; / ; _ ; \overset{\{ \} {a}}{} \$\$\$

R35B — Degemination of [II]

\$\$ \text{II} \rightarrow I \$\$\$

R36A — Loss of [w] in [Vwa] when no vowel is stressed

\$\$ w \rightarrow \emptyset ; / \text{!} / ; \begin{bmatrix} V \backslash -\text{acc} \end{bmatrix} _ \begin{bmatrix} V \backslash +\text{low} \backslash -\text{acc} \end{bmatrix} \$\$\$

R36B — Lengthening of f.sg. definite article [a]

\$\$ a \rightarrow [+ \text{long}] ; / ; _ ; \# \quad \text{(f sg n/a articulated forms)} \$\$\$

R36C — [uw] → [uv]

\$\$ w \rightarrow v ; / ; u _ a \$\$\$

R37A — Raising of non-initial, non-geminate unstressed [a]

\$\$ \begin{bmatrix} V \backslash +\text{low} \backslash -\text{acc} \backslash -\text{long} \end{bmatrix} \rightarrow [-\text{low}] ; / ; C _ \$\$\$
\$\$ a \rightarrow \text{partial} ; \text{(unstressed, non-initial)} \$\$\$

R37B — Degemination of geminate vowels

\$\$ V_i ; V_i \rightarrow V_i \$\$\$

R38 — Desyllabification of [e] before [a]

\$\$ e \rightarrow [-\text{syl}] ; / ; _ ; a \$\$\$

R39 — Lowering of unstressed internal [i]

\$\$ \begin{bmatrix} V \backslash +\text{high} \backslash -\text{back} \backslash -\text{acc} \end{bmatrix} \rightarrow [-\text{high}] ; / ; C _ C \$\$\$
\$\$ i \rightarrow e ; / ; [-\text{acc}] \$\$\$

R40 — Diphthongization of [i]: insertion of [j]

\$\$ \emptyset \rightarrow j ; / ; _ ; i \$\$

\$\$ i \rightarrow ji \$\$

R41A — Palatalization of non-occlusive alveolars before [j]

\$\$ \begin{bmatrix} +\text{cont} \backslash +\text{ant} \backslash +\text{cor} \end{bmatrix} \rightarrow \begin{bmatrix} +\text{high} \backslash -\text{ant} \end{bmatrix} ; / ; _ ; j \$\$

\$\$ s \rightarrow \check{s}, \quad l \rightarrow \lambda, \quad n \rightarrow \tilde{n} ; / ; _ ; j \$\$

R41B — Absorption of [j] after palatal

\$\$ j \rightarrow \emptyset ; / ; [+ \text{high}] _ \$\$

R42A — Affrication of alveolar stops before [j]

\$\$ \begin{bmatrix} -\text{son} \backslash -\text{cont} \backslash +\text{ant} \backslash +\text{cor} \backslash -\text{del;rel} \end{bmatrix} \rightarrow [+ \text{del;rel}] ; / ; _ ; j \$\$

\$\$ t \rightarrow ts, \quad d \rightarrow dz ; / ; _ ; j \$\$

R42B — Absorption of [j] after affricate

\$\$ j \rightarrow \emptyset ; / ; [+ \text{del;rel}] _ \$\$

R43 — Palatalization of [l] after velar

\$\$ [+ \text{lat}] \rightarrow \begin{bmatrix} +\text{high} \backslash -\text{ant} \end{bmatrix} ; / ; \begin{bmatrix} -\text{cont} \backslash -\text{ant} \backslash -\text{cor} \end{bmatrix} _ \$\$

\$\$ l \rightarrow \lambda ; / ; k, g _ \$\$

R44 — Fronting of [ə] after [ʌ] and [ɯ]

\$\$ \begin{bmatrix} V \backslash +\text{back} \end{bmatrix} \rightarrow [-\text{back}] ; / ; \begin{bmatrix} -\text{cont} \backslash +\text{high} \backslash -\text{strid} \end{bmatrix} _ \$\$

\$\$ \partial \rightarrow e ; / ; \lambda, \tilde{n} _ \$\$

R45A — Assimilation [s] → [š] before [š]

\$\$ s \rightarrow [-\text{ant}] ; / ; _ \begin{bmatrix} +\text{del;rel} \end{bmatrix} \backslash -\text{ant} \end{bmatrix} \$\$

R45B — Dissimilation [š] → [t] after [š]

\$\$ \begin{bmatrix} +\text{del;rel} \end{bmatrix} \backslash -\text{ant} \end{bmatrix} \rightarrow [-\text{del;rel}] ; / ; \check{s} _ \$\$

Daco-Romanian Period (11th century onward)

R46A — Backing of [e]/[ɛ] after labial before back-vowel syllable

\$\$ \begin{bmatrix} V \backslash -\text{cons} \end{bmatrix} \backslash -\text{back} \end{bmatrix} \rightarrow [+ \text{back}] ; / ; \begin{bmatrix} C \backslash +\text{ant} \end{bmatrix} \backslash -\text{cor} \end{bmatrix} _ ; C^0 \begin{bmatrix} V \backslash +\text{back} \end{bmatrix} \end{bmatrix} \$\$

\$\$ e, \text{varepsilon} \rightarrow \text{partial}, a ; / ; \text{labial} _ ; \cdots ; V_{[+ \text{back}]} \$\$

R46B — Loss of [ə] before [a]: [əa] → [a]

\$\$ \text{partial} \rightarrow \emptyset ; / ; _ ; a \$\$

R47A — Raising of [ə] before [n] or [mCV]

\$\$ \begin{bmatrix} V \backslash +\text{low} \end{bmatrix} \rightarrow [+ \text{high}] ; / ; _ \begin{Bmatrix} n \backslash \text{mCV} \end{Bmatrix} \end{Bmatrix} \$\$

\$\$ a \rightarrow \text{hat{i}} ; / ; _ ; n(C) \$\$

R47B — Raising of [e] before nasal

\$\$ \begin{bmatrix} V \backslash -\text{high} \end{bmatrix} \backslash -\text{back} \end{bmatrix} \rightarrow [+ \text{high}] ; / ; _ \begin{bmatrix} C \backslash +\text{nas} \end{bmatrix} \end{bmatrix} \$\$

\$\$ e \rightarrow i ; / ; _ ; n, m \$\$

R48 — Degemination of [nn]

\$\$ nn \rightarrow n \$\$

R49 — Raising of [ɛ] before [ɛ] or [ɛ] plus consonant

R49 — Raising of [ə] before [r] or [s] plus consonant

$$\partial \rightarrow \hat{i} ; / ; _ \begin{Bmatrix} r \\ s \end{Bmatrix} C$$

R50 — Backing of front vowels after [rr]

$$\begin{bmatrix} V \\ -\text{back} \end{bmatrix} \rightarrow [+ \text{back}] ; / ; [+ \text{rep}] _$$

R51 — Simplification of [rr] → [r]

$$[+ \text{rep}] \rightarrow [- \text{rep}]$$

R52A — Deaffrication of voiced alveolar affricate

$$\begin{bmatrix} +\text{del;rel} \\ +\text{ant} \\ +\text{voice} \end{bmatrix} \rightarrow [-\text{del;rel}]$$

$$dz \rightarrow z$$

R52B — Deaffrication of voiced palatal affricate before [o] or [u]

$$\begin{bmatrix} +\text{del;rel} \\ -\text{ant} \\ +\text{voice} \end{bmatrix} \rightarrow [-\text{del;rel}] ; / ; _ \begin{bmatrix} V \\ +\text{round} \end{bmatrix}$$

$$\hat{j} \rightarrow \check{z} ; / ; _ ; o, u$$

R53A — Monophthongization of [ea] after [š] or [ts]

$$ea \rightarrow a ; / ; \begin{Bmatrix} \check{s} \\ ts \end{Bmatrix} _ \quad \text{(sporadic)}$$

R53B — Monophthongization of [ea] before a syllable containing [e]

$$ea \rightarrow e ; / ; _ ; C^0 ; e$$

R54 — Delateralization of [λ]

$$\begin{bmatrix} +\text{lat} \\ +\text{high} \\ -\text{ant} \end{bmatrix} \rightarrow \begin{bmatrix} -\text{lat} \\ -\text{cons} \\ -\text{syl} \end{bmatrix}$$

$$\lambda \rightarrow j$$

R55 — Denasalization of [ɲ]

$$\begin{matrix} \text{son} \\ \text{ant} \\ \text{cor} \end{matrix} \rightarrow \begin{matrix} \text{nas} \\ \text{cons} \\ \text{lat} \end{matrix}$$

$$\tilde{n} \rightarrow j$$

R56 — Loss of [j] before [i]

$$j \rightarrow \emptyset ; / ; _ ; i$$

R57 — [jea] → [ja]

$$e \rightarrow \emptyset ; / ; j _ a$$

R58 — Word-final [ə] → [e] after [i]

$$V \rightarrow [-\text{back}] ; / ; i _ ; \#$$

R59A — Devoicing of word-final [u] and [i]

$$V \rightarrow [-\text{high}] \rightarrow [-\text{voice}] ; / ; X _ ; \#$$

where $X \neq [\text{consonant} + \text{liquid}]$

R59B — Loss of word-final voiceless [w]

$$\begin{matrix} \text{cons} \\ \text{syl} \\ \text{voice} \end{matrix} \rightarrow \emptyset ; / ; _ ; \#$$

R59C — [ii] → [i] word-finally

$$i ; i \rightarrow i ; / ; _ ; \#$$

R60 — Loss of final [l] of masculine singular definite article

$$l \rightarrow \emptyset ; / ; _ ; \# \quad \text{(m sg definite article only)}$$

A few notational notes: the symbol ∂ is used throughout for Romanian [ə/i] (the central

unrounded vowel written $\langle \hat{a}/\hat{i} \rangle$, $\hat{i}$ for the same vowel in raised environments, λ for the palatal lateral [ɭ], $\tilde{n}$ for palatal nasal [ɲ], β for the bilabial fricative, and ε for lax [ɛ]. Rules marked *sporadic* are explicitly flagged as such in the source. The feeding/bleeding relationships Pardess establishes (especially among R27A, R27B, R31A, R31B, and the palatalization rules R41–R42) are encoded implicitly in the ordering above.

User: what does "-acc" mean?

use.ai: In SPE notation, $[-acc]$ means **minus the feature "accent"** — i.e., the segment or syllable is **unstressed**.

The feature $[acc]$ (or $[+acc]$) marks the **primary stressed** syllable. So:

$[+acc]$ = bears primary stress

$[-acc]$ = lacks primary stress (unstressed)

In Pardess's own abbreviation list, he writes $[acc]$ as standing for **"primary stress"**, which he treats as a binary distinctive feature on vowels, following the generative practice of encoding stress as a feature rather than as a separate suprasegmental tier (as later theories like metrical phonology would do).

So a rule like R28:

$$\begin{matrix} V \\ +\text{back} \\ -\text{high} \\ -\text{low} \\ -\text{acc} \end{matrix} \rightarrow [+ \text{high}]$$

reads: **an unstressed, non-high, non-low back vowel raises** — i.e., unstressed [o] becomes [u].